BIRZEIT UNIVERSITY

Faculty of Engineering and Technology
Computer Science Department
Human Computer Interaction (COMP322)

Phase 1 Report

Birzeit library catalogue and e-resources

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1.Team and Topic

This report is prepared by **Group 5**: Hanan Naji (1220651), Rahaf Badwan (1220654), Alaa Tanineh (1221601), and Ayah Sarama (1221615). The team members divided the work across the main Phase 1 tasks, including heuristic evaluation, exploratory interviews, personas, scenarios, and usability requirements. The chosen topic is the **Birzeit University Library catalogue and e-resources**, accessed through the university library website.

The scope of this project focuses on how students discover, locate, and access physical and electronic library resources. Specifically, we evaluate the catalogue search interface and the e-resource access paths, such as online databases, e-journals, and off-campus access links. The redesign scope is limited to three main user journeys:

- known-item search, filtered exploratory search, and off-campus e-resource access. In the known-item search flow, users search for a specific book, article, title, author, or course-related resource.
- In the filtered exploratory search flow, users explore resources by subject, language, availability, or online access.
- In the off-campus access flow, users try to reach electronic resources while studying remotely.

This problem matters because the library system supports essential academic activities such as finding course readings, preparing assignments, accessing research materials, and studying outside campus. If the catalogue or e-resource paths are unclear, students may waste time, fail to find available resources, or assume that the library does not provide the materials they need. Therefore, Phase 1 focuses on understanding current user needs and usability problems through heuristic evaluation, interviews, personas, scenarios, and measurable usability requirements.

2. Heuristic evaluation (Note: The screenshots referenced in the "# Figure" column is provided in the Appendix.)

ID#	# Figure	System Area / Screen	Heuristic Violated (1–10)	Description of the Problem	Frequency / Impact / Persistence	Severity (0–4)	Suggested Fix
1	3	**Item Reservation**	H1 – Visibility of System Status	When students attempt to reserve a library item, the system redirects them directly to the login page without informing them beforehand that authentication is required. Users may not understand why they were redirected or whether the reservation process has started successfully.	**Frequent / Major / Persists**	3 – Major usability problem. Students frequently attempt to reserve library materials. The system redirects them immediately without prior notice, interrupting task flow and confusing first-time users.	Implement a Login Requirement Notification before redirecting users. Display a clear message such as "Login required to reserve this item" so users understand why authentication is needed.
2	4	**Catalogue Search**	H1 – Visibility of System Status	When the user starts a search in the library catalogue, the system does not show a clear loading indicator or progress feedback. The user cannot easily tell whether the search is processing, delayed, or whether the click was registered.	**Frequent / Minor / Persists**	2 – Minor usability problem. This does not completely prevent the user from searching, but it may cause confusion and repeated clicks, especially if the search takes time to load.	Add a visible loading spinner or message such as "Searching…" immediately after the user starts a search. After processing, show either the results or a clear "No results found" message.
3	6	**Library Homepage**	H6 – Recognition Rather Than Recall	The main search entry is not clearly visible on the page. First-time users may not realize that must infer that the magnifying-glass icon opens the catalogue search to access the correct search box. The page contains several visual elements and links that compete for attention.	**Frequent / Minor / Easy to overcome**	2 – Minor usability problem. The search function is still available, but first-time users may take extra time to recognize that they must click the magnifying-glass icon. This slows down the search process.	Make the search entry clearly visible by showing a direct search box with a clear label such as "Search the library catalogue." Add a visible Search button at the top of the page.
4	4	**Catalogue Search**	H9 – Help Users Recognize, Diagnose, and Recover from Errors	The catalogue search requires very exact spelling. For example, if the user searches for "java1" instead of "java," the system shows no results even though many Java books are available. The system does not suggest similar keywords or correct the typo.	**Frequent / Major / Persists**	3 – Major usability problem. If small typing mistakes lead to zero results, users may incorrectly believe that the library does not have the needed resources.	Add typo tolerance and search suggestions. The system should show messages such as "No exact results found for 'java1'. Did you mean 'java'?"
5	4	**Catalogue Homepage**	H4 – Consistency and Standards	The catalogue page mixes Arabic and English labels within the same interface. The search box says	**Frequent / Minor /**	2 – Minor usability problem. Users may still understand the	Use one consistent interface language across the page, or

				"Search the catalog" and the dropdown says "Keyword," while the navigation links and page title are in Arabic. This inconsistency makes the interface less coherent.	**Persists**	page, but the mixed language reduces consistency and can confuse users who expect a fully Arabic or fully English interface.	provide a clear language switch.
6	1	**Search Field Selector**	H7 – Flexibility and Efficiency of Use	The search function relies on manual text input combined with a field type selector (Title, Author, Subject, ISBN, ISSN, Series, Call Number). Students often do not know which field type to choose, and the system provides no guidance, examples, or tooltips.	**Frequent / Minor / Persists**	2 – Minor usability problem. The system remains functional but the lack of guidance increases cognitive effort for inexperienced users without fully blocking task completion.	Add illustrative examples beneath the search box for each field type. Support real-time autocomplete suggestions while typing.
7	1	**Search Results Page**	H8 – Aesthetic and Minimalist Design	The results page is visually overloaded. Each result simultaneously displays: title, author, edition, material type, language, publisher, call number, availability, cover image, star rating, Reserve button, Save to Lists, Add to Cart, and a checkbox. A top toolbar adds further actions.	**Frequent / Minor / Persists**	2 – Minor usability problem. The visual clutter is disruptive and slows down the user, but it does not prevent finding results. The problem increases scanning time and mental load.	Simplify each result card to show only: title, author, year, and availability. Group secondary actions into a collapsible "More Options" menu.
8	1	**Navigation Bar – Advanced / Citation / Browse Search**	H2 – Match Between System and the Real World	The navigation bar contains three options unfamiliar to average students: Advanced Search (Boolean operators), Citation Search (technical terms like "Full Record Limits"), and Browse Search (no explanation of difference from regular search). None include tooltips or usage examples.	**Frequent / Major / Persists**	3 – Major usability problem. Students avoid these options entirely and fall back to basic keyword search, leaving important search features completely unused.	Add a short description at the top of each page explaining when and how to use it. Replace technical labels with plain, student-friendly language.
9	5	**Search Field Selection**	H6 – Recognition Rather Than Recall	The system provides multiple search options (Keyword, Title, Author, Subject, ISBN, Call Number) but does not explain which to choose when a term has multiple academic meanings. A student searching for "Python" may not know whether to use Keyword, Subject, or Title.	**Frequent / Major / Persists**	3 – Major usability problem. Students must rely on prior knowledge or trial and error to choose the correct search type, increasing cognitive load and search time.	Add tooltips or example queries next to each category. For example: "Keyword: Search any related term", "Subject: Search by academic field", "Title: Search exact book titles".
10	2	**Library Homepage – Off-Campus Access**	H6 – Recognition Rather Than Recall	The system separates the main library search interface from the off-campus access feature across different visual areas. The interface does not clearly communicate their connection, requiring users to remember where remote access is located.	**Frequent / Major / Persists**	3 – Major usability problem. Students who need off-campus access must remember where the feature is located. This reduces discoverability, especially for first-time users.	Integrate the off-campus access option closer to the main search section, or add a clearly labelled button such as "Off-Campus Access" beside the main search entry point.

2.1 Summary:

The heuristic evaluation revealed that most usability issues are related to discoverability, unclear terminology, and insufficient guidance for novice users. The most severe problems were associated with search functionality, off-campus access, and system feedback during authentication and error recovery, because these issues directly interrupt core academic tasks. Students may fail to find required resources, misunderstand access permissions, or abandon the system entirely, especially first-time users who rely heavily on clear guidance and feedback.

3. Exploratory Interviews:

3.1 Methods

We did three interviews to see how people use the Birzeit Library Catalogue and E-Resources. The people we talked to were a first-year student, a fourth-year student and someone who works at the library. We wanted to know three things: when the Birzeit Library Catalogue and E-Resources users look for something they know when they use filters to look around and when they try to get to the Birzeit Library Catalogue and E-Resources from home.

Each interview took 20 to 30 minutes. Before we started each person signed a form to say it was okay. We did not use the Birzeit Library Catalogue and E-Resources users names instead we used codes like P1, P2 and P3, for the Birzeit Library Catalogue and E-Resources users.

3.2 participants

Code	Participant profile	Device used	Relevance to study
P1	First-year Computer Engineering student	Laptop + mobile	New user with limited experience using the library website
P2	Fourth-year Computer Science student	Laptop + mobile	Experienced user who has used the library website for projects, research, electronic articles, filters, and off-campus access
P3	IT Unit Staff / Library System Support Participant Data	Laptop + mobile	System expert involved in supporting and troubleshooting the library system, with insights into user access, search, and usability problems.

3.3 key findings

Finding 1: New students do not know where to start

The library website is a bit of a mystery to students like P1. He said that when he needs to find something he usually starts with Google because it is easier and faster for him.

Finding 2: Availability information is very important but not clear

P1 thinks that knowing if something is available is really important. If the source is not available then it is not useful to him. He only understands that something is available online if it says "online" clearly.

Finding 3: Filters are known in general. Not used confidently

P1 knows what filters are. He is not really sure when to use them. He does not know when to use the subject, language or availability filters, in the library system. P3 also confirmed that most users rarely use filters because they do not understand their purpose or how they improve search results.

Finding 4: Off-campus access is confusing

P1 tried to access something from home one time. He got confused. The system took him to pages and he was not sure if he was allowed to access what he needed. P3 also highlighted that users struggle with off-campus access because the "Access databases from outside the University" option is not clearly located in the system, causing confusion about login and access.

Finding 5: Users rely on workarounds

When the library system is not clear P1 uses Google. Asks his friends or older students for help instead of trying to figure out the library website.

Finding 6: Users struggle with understanding search results structure

P3 explained that users often do not understand how search results are organized. They get too many results without clear structure or explanation, which makes it difficult for them to identify the correct item.

Finding 7: System terms is too technical for users

According to P3, many terms used in the system are too technical for average users. New students especially find it hard to understand labels and categories inside the library system.

Finding 8: Lack of guidance and onboarding for new users

P3 emphasized that there is no simple guide or onboarding support inside the system. This leads users, especially new ones, to feel lost and rely on trial-and-error or external help.

3.4 Summary

Interview evidence was taken into account when developing personas, scenarios, and usability requirements. Persona P1 described requirements for the design for new users: a definite entry point, easy-to-understand terminology, visible availability indicators, and assistance in using the system off campus. Persona P2 described requirements for the design for users with experience using library services: structured search results, better use of filters, and clearer distinction between printed books, digital resources, and external databases. Finally, persona P3 reiterated commonly occurring user problems: unclear searching, vague results, poor filter performance, unclear logins, and inability to access digital materials from outside of the university.

4. Personas

Persona 1 — Primary Persona

Field	Your persona (fill in)
Persona name	Ahmad Khalil
Photo or sketch	
Demographics	First-year Computer Engineering student at Birzeit University. Native Arabic speaker. Uses a laptop for assignments and a mobile phone for quick checks.
Tech literacy	Medium for everyday apps, but low confidence with academic library systems. Comfortable with Google, WhatsApp, and mobile apps, but not confident with filters, catalogue terminology, or terms such as E-Resources. Prefers Arabic interfaces and simple navigation

Goals (2-3)	• Find required course books and materials quickly. • Know clearly whether a source is available online or physically in the library. • Reserve books and access resources from home with clear login instructions.
Frustrations	• Does not know where to start and usually starts with Google because the library website feels unclear. • Gets confused by filters, search results, and terms such as E-Resources. • Gets lost during off-campus access and reservation because the system does not clearly explain which account to use or whether the action succeeded. • Uses Google or asks friends when the library system is unclear.
Context of use	Usually uses the system from home while studying or completing assignments. Uses a laptop for research tasks and a mobile phone for quick searches on campus. When login, reservation, or access steps are unclear, he asks friends for help.
Representative quote	"Google is easier and faster, and I am not exactly sure how to search in the library website. It feels like it needs a lot of explanation. "Source: P1
Evidence trail	This persona emerged from the beginner / low-confidence searcher theme. It is anchored by P1 and supported by P3's observations about recurring problems among new users. Participant codes: P1, P3. Themes: Google-first searching, filter confusion, E-Resources confusion, unclear off-campus access, unclear reservation login, reliance on friends, need for simple onboarding.
Behavior patterns	• Starts with Google before using the library system. • Rarely uses filters. • Stops searching if results become confusing. • Depends on trial-and-error and peer support.
Needs from the redesigned system	• Clear and visible search entry point. • Simple explanations and onboarding guidance for first-time users. • Clear labels showing whether a source is available online or physically. • Easy off-campus login instruction explaining which account to use

Persona 2 — Secondary Persona

Field	Your persona (fill in)
Persona name	Lina Odeh
Photo or sketch	
Demographics	Fourth-year Computer Science student at Birzeit University. Frequently works on projects and academic research. Comfortable using Arabic and English for academic searching.
Tech literacy	High. Frequently starts with Google Scholar and sometimes uses IEEE or ACM. Uses a laptop for research and a mobile phone only for quick checks. Understands filters and academic search concepts better than beginner students, but still needs clearer paths for databases, e-resources, and off-campus access.
Goals (2-3)	• Access reliable academic papers and electronic resources efficiently. • Narrow large search results using filters such as title, author, source type, year, and availability. • Access library databases remotely with clear login and availability information.
Frustrations	• Off-campus access and external database login pages make it unclear whether to use the university account or another account. • The system feels split between catalogue, databases, and electronic resources, so it is

	not always clear where to start or return. • Reservation confirmation is not clear enough after pressing the reservation button. • Filters are useful, but results are not always understandable from the first attempt.
Context of use	Uses the library website both from home and on campus, mainly while working on projects or assignments. Uses laptop for long research sessions and mobile only for quick checks.
Representative quote	"Sometimes I am not sure if the login page belongs to the university or to an external database." Source: P2
Evidence trail	This persona emerged from the experienced / research-focused user theme. It is anchored by P2 and supported by P3's observations about recurring problems with search organization, database access, technical terminology, reservation feedback, and off-campus login. Participant codes: P2, P3. Themes: Google Scholar searching, use of filters, comparing title/author/year, database access, off-campus login confusion, unclear reservation feedback, need for clearer resource paths.
Behaviour patterns	• Uses filters and advanced search more than beginners. • Reads search details carefully before selecting resources. • Uses Google Scholar and external databases alongside the library system. • Tries troubleshooting problems independently before asking for help.
Needs from the redesigned system	• Better organized search results and clearer filtering. • Step-by-step off-campus access guidance explaining which account to use. • Clear distinction between catalogue results, databases, e-resources, physical books, and full-text articles. • Clear reservation confirmation after pressing the reservation button.

5. Scenario

Scenario 1 — Known-item search and reservation

Omar, a first-year Computer Engineering student, is working on an assignment from home. His instructor gives him the title of a required book, but Omar is not confident using the library website and usually starts with Google. In the redesigned system, he opens the library homepage and immediately sees a clear search box labelled "Search the library catalogue." He types the book title, finds the correct result, and clearly sees whether it is available in the library or online. When he clicks "Reserve," the system explains that login is required before redirecting him. After logging in with his university account, Omar receives a clear confirmation message with the book title and pickup information.

Scenario 2 — Filtered exploratory search

Lina, a fourth-year Computer Science student, is preparing a research project from home using her laptop. She searches for sources about artificial intelligence in education and gets many results. In the redesigned system, the results page clearly separates books, articles, and online resources. Lina uses simple filters such as source type, year, availability, and online access to narrow the results. Short explanations beside the filters help her choose the right options without guessing. She opens a recent article, checks the title, author, year, and full-text availability, then saves it for her project.

Scenario 3 — Off-campus e-resource access

Lina is working late at night from home and finds an electronic article that may help her project. In the old system, she often felt unsure which login account to use or whether the article was actually available through the university. In the redesigned system, the article page shows a clear "Access from off

campus" option. Before redirecting her, the system explains that she should use her university account and tells her what to expect after login. Lina logs in once, opens the full text successfully, and sees a confirmation that access is active through Birzeit University. If the article is not available, the system gives a clear message instead of leaving her confused.

6. Measurable usability requirements

The requirements listed below were derived based on heuristic evaluation, exploratory interviews, persona development, and scenario development for the proposed redesign of the catalogue and electronic resource systems of Birzeit University Library. Following the usability engineering process, all requirements are written as a measurable target rather than a vague design goal. The requirements have been mapped against ISO 9241-11 usability goals: effectiveness, efficiency, and satisfaction.

UR1: Make search easy to find
The new homepage should have a search box for the library catalogue. This will help users find what they need fast. Most users, least 90% should be able to start searching for something they know exists within 10 seconds on their own.

UR2: Show if resources are available
The system should clearly state if a resource is available in the library borrowed, available online or not available. This helps users. Least 90% of users should know if a resource is available or not without having to click further.

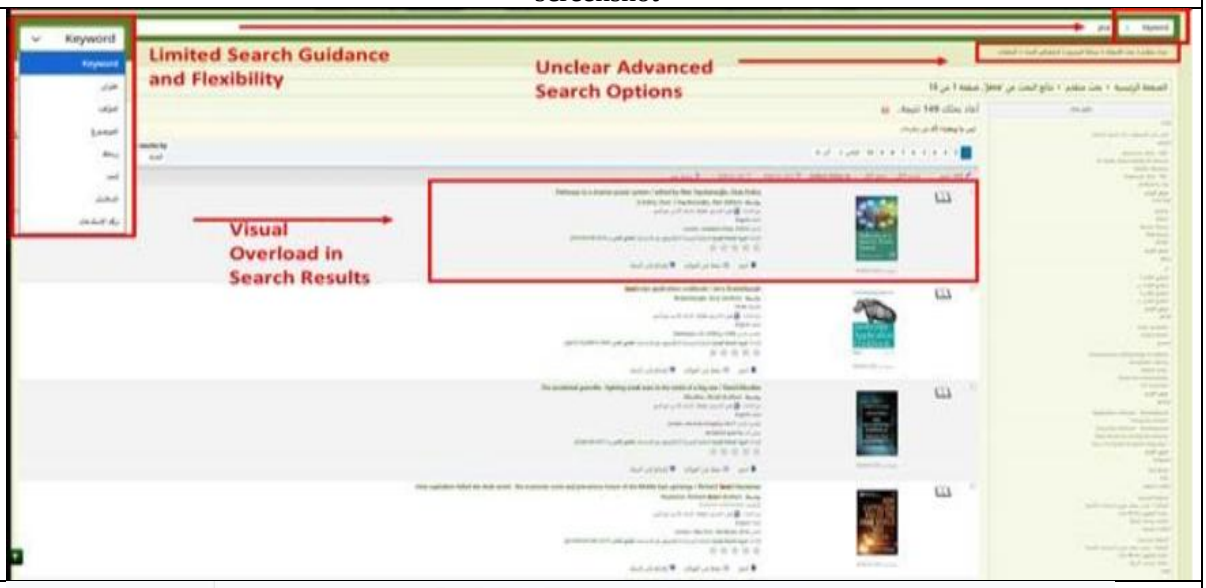
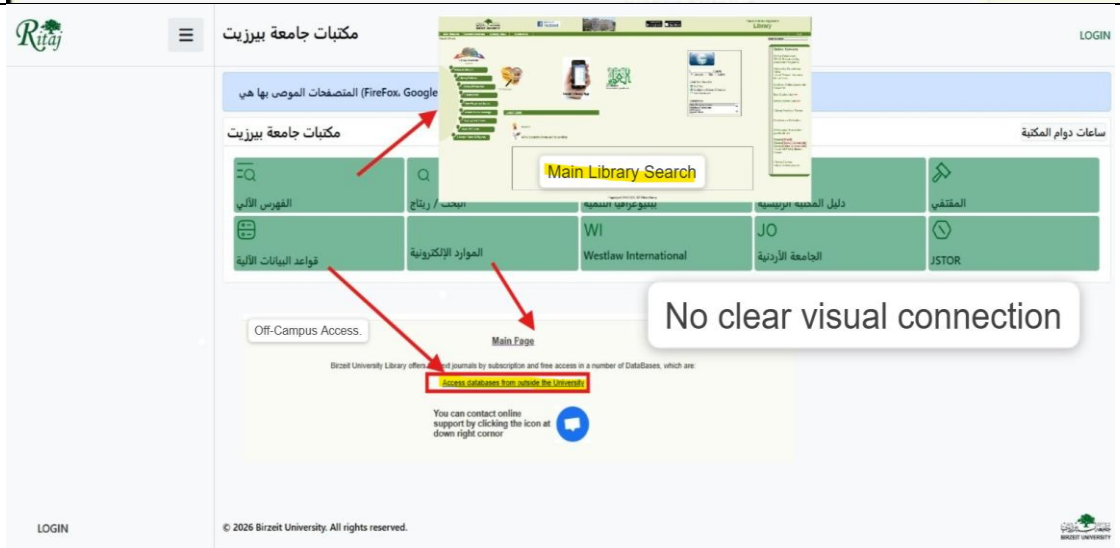
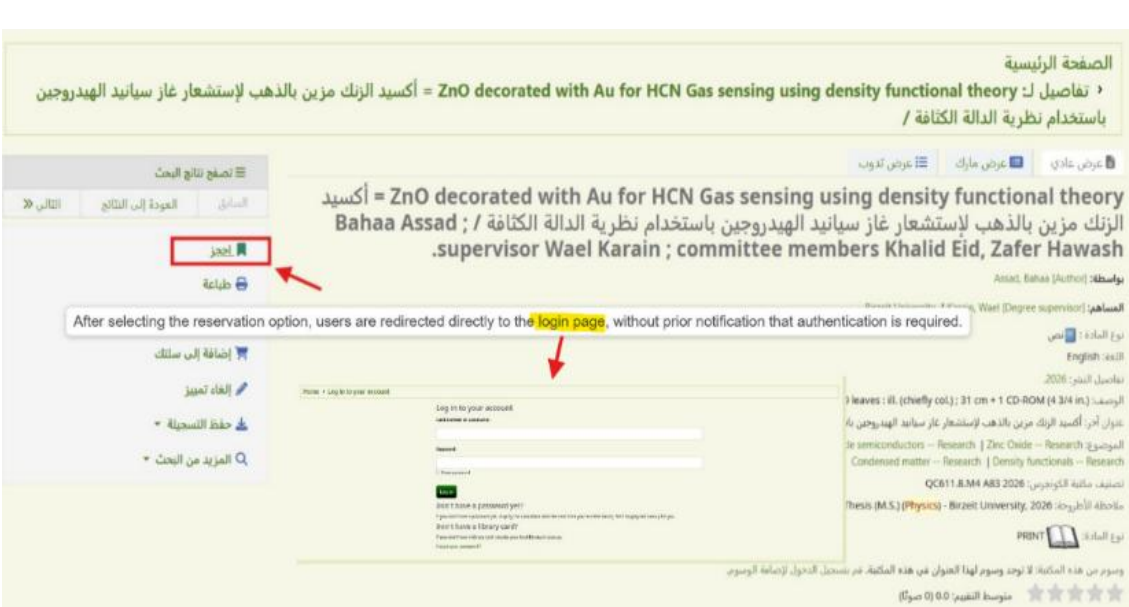
UR3: Help with search mistakes
The search function should be forgiving if users make spelling mistakes. It should suggest options if it can't find anything. This makes searching easier and more satisfying. At 80% of users should be able to find what they're looking for within 60 seconds if they made a mistake.

UR4: Make search results easy to read
The search results page should show the important details first: title, author, year, what type of resource it is and if its available. This helps users find what they need quickly and makes them happy. Least 80% of users should find the right result within 2 minutes and think it's clear rating it 4 out of 5.

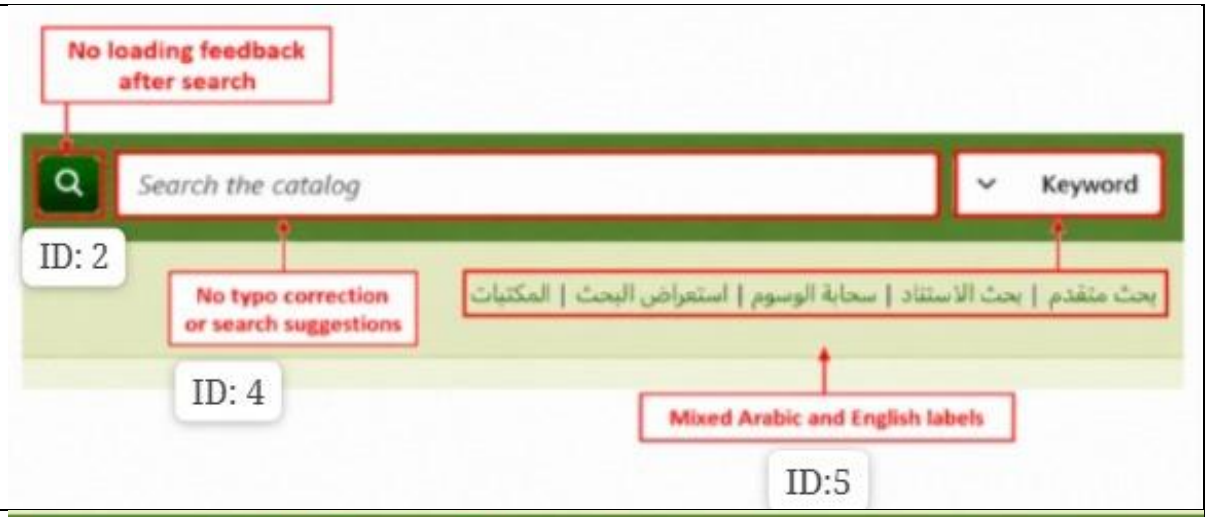
UR5: Easy off-campus access
The option to access from campus should be easy to find. It should explain what to do before taking users to a login page or another database. This makes it easier and more satisfying for users. At 85% of users off-campus should be able to get in without help and 80% should think the process is clear.

UR6: Give feedback away
The system should give clear messages for important actions, like searching reserving an item or logging in. This helps users. Makes them happy. All these actions should give feedback in 1 second like "Searching…" "You need to login" or "Redirecting to database."

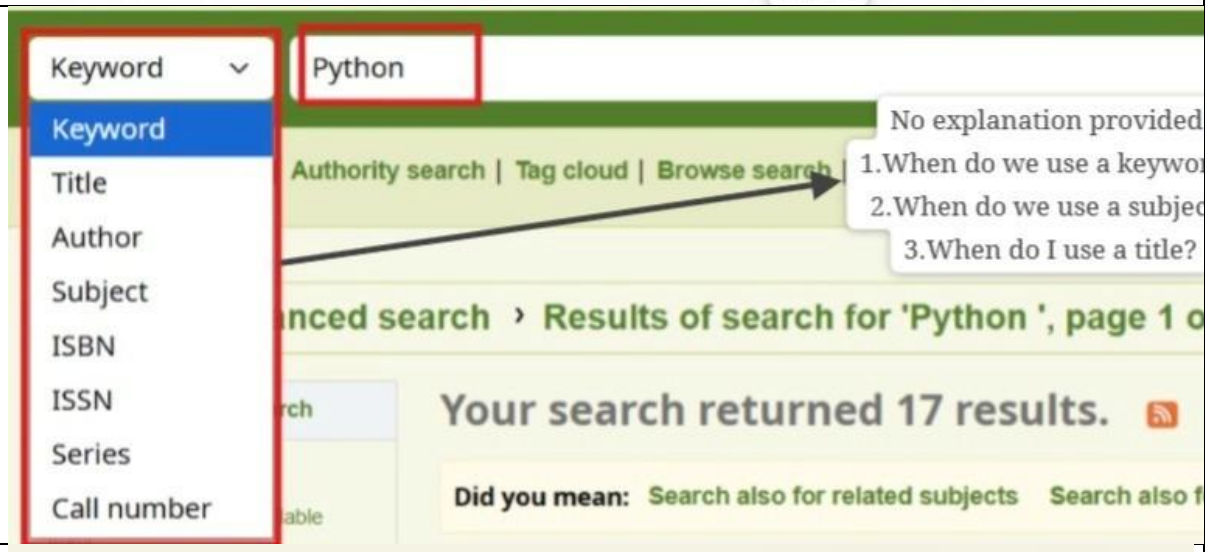
Appendix

#	screenshot
1	
2	
3	

4


No loading feedback after search
Search the catalog
Keyword
ID: 2
No typo correction or search suggestions
ID: 4
بحث متقدم | بحث الاستناد | سحابة الوسوم | استعراض البحث | المكتبات
Mixed Arabic and English labels
ID:5

5


Keyword
Python
Keyword
Title
Author
Subject
ISBN
ISSN
Series
Call number
Authority search | Tag cloud | Browse search
No explanation provided
1.When do we use a keywo
2.When do we use a subjec
3.When do I use a title?
nced search › Results of search for 'Python ', page 1 o
Your search returned 17 results.
Did you mean: Search also for related subjects

6


إبحث عن كتاب، مجلة، مقال ...
الفهرس الآلي
Search entry is unclear; users must click the magnifying-glass icon to start searching.
الفهرس القديم

Appendix

- **Appendix A: Signed Consent Forms**

  A1. Consent Form for Person 1:
  https://docs.google.com/document/d/1sgtJBcM_UDNuqqnZka9XjV2Jgi_J2qWT/edit?usp=drive_link&ouid=106707978615167847619&rtpof=true&sd=true

  A2. Consent Form for Person 2:
  https://drive.google.com/file/d/1EEQUN4RBDYIRhCEoJqzJ2zsNv37TO3vb/view?usp=drive_link

  A3. Consent Form for Person 3: https://docs.google.com/document/d/1H7-ZEB0G-bh68H22RpqR4nS3x6FB-qGC/edit?usp=drive_link&ouid=106707978615167847619&rtpof=true&sd=true

- **Appendix B: Interview Guide Used**

  B1. The questions we asked during the interviews:
  https://docs.google.com/document/d/1E4y8ts09-OD675R6d0JEatoLbsyKSJr6/edit?usp=drive_link&ouid=106707978615167847619&rtpof=true&sd=true

  Appendix C: Interview Notes

  C1. What Person 1 said during the interview
  https://docs.google.com/document/d/1fuRQcQaJ9pIuk3byMB89u8xKeQaCHjzL/edit?usp=drive_link&ouid=106707978615167847619&rtpof=true&sd=true

  C2. What Person 2 said during the interview:
  https://drive.google.com/file/d/1eHEJLXzlrminoW7UVOT_HyStu3iA1oZs/view?usp=drive_link

  C3. What Person 3 said during the interview
  https://docs.google.com/document/d/1IiNveeWkc2kjagBzyKQQiBqzkuBP_8Zu/edit?usp=drive_link&ouid=106707978615167847619&rtpof=true&sd=true

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1. Conceptual Model

The redesigned Birzeit University Library system is based on an idea that makes it simple and easy for people to access and use academic resources. It does not require users to know library terms. Instead, it follows what students normally do when they look for information: they search for something look at the details and then use it or reserve it.
The system has three parts:

Stage	User Action	System Response
Search	Enter a word, title or author name (type what the user need)	Show clear results that match and if they are available
Evaluate	Look at the details of a resource means review details	Show information, like title, author, year and what kind of resource it is
Access / Reserve	Use or reserve the resource	Guide the user log in confirm and get access

This concept model specifically considers the key issues discovered in Phase 1. As Ahmad in Phase 1 was having difficulty identifying a start-point, the model first includes the Search process stage. As both P1 and P3 had issues regarding the availability of information resources, the Evaluate process stage takes this into account and identifies availability and further information prior to any action being taken. Reservation and off campus access problems raised by P1, P2, and P3 have been solved in the Access/Reserve process stage.

The proposed conceptual model is created based on the key issues raised in Phase 1. In this regard, the redesigned system offers one obvious entry point for finding something, allows to evaluate findings through a simple filtering process, detailed resource descriptions, and availability indicators. Also, it allows to work with both physical and electronic resources.

helping users to either reserve one or access it electronically. In case when login or off-campus access is needed, the user will receive appropriate information beforehand, and get feedback upon completion of their request.

This means that the system will operate with two types of resources: physical, which should be found within the library building using location data; and electronic, which are accessible through the Web directly. Therefore, users will have an opportunity to search for something, filter findings, view details, make reservations, and access resources all from one place.

2. How the design addresses each persona

2.1 Ahmad Khalil

Ahmad is afirst year in the Computer Engineering department with little experience interacting with library systems. He mostly relies on the search results from Google because he finds most library website interfaces very complicated and difficult to work through. The new redesigned system is benefet for Ahmad because the search bar appears in clear view on the homepage allowing him to quickly begin his searches. Whether the material is available in the library or online is readily available for Ahmad through clear icons ,color and tags. The

system gives him explicit steps on what information is required for authentication prior to redirection; reducing uncertainty and confusion for Ahmad. Ahmad can now recover from typos easily through suggestions.

These design decisions directly address Ahmad's main frustrations identified in Phase 1, including uncertainty about where to start searching, confusion about availability information, and difficulty accessing resources from home.

2.2 Lina Odeh

Lina is a fourth year in the Computer Science department and regularly accesses academic resources for research.she is a library system user and needs easy and efficent search tools and remote access. The new redesigned system supports advanced filters for her to refine his searches based on resource type, availability and publication year.
Resources are clearly marked as to type, with labels and icons, so you can tell at a glance whether it's a book, an article or an electronic resource. The design also includes a defined process for off-campus access with clear instructions step-by-step. Finally, reservation confirmations and feedback messages ensure that Lina always knows the status of her action

These improvements support Lina's need for efficient research workflows, clearer resource organization, and reliable off-campus access, which were recurring themes identified during the interviews.

3. Design Decisions Based on Norman's Principles

This redesign attempts to solve the existing problem by implementing the design principles recommended by Norman. Specifically, the design solutions are meant to address the issues identified during Phase 1, such as unclear search starting point, unclear reservation process, absence of feedback, and difficulties in accessing off-campus resources.

Visibility: First, the new Home Page makes the primary search function easy to find, This decision directly addresses Finding 1 from Phase 1, where P1 reported uncertainty about where to begin searching. All essential options, including Search, E-Resources, filters, and Off-Campus Access, are presented clearly in the prototype instead of being buried under a complex menu structure.

Feedback: This principle is applied by providing clear feedback after key user actions. In particular, this is true for search operations, since the user receives feedback while the search results are still being loaded, and during the reservation process, where the Login Required page and Reservation Confirmation page guide the user, as shown in **Figures A2, A5, and A7**.

This design decision addresses both Heuristic Issue ID1 and Finding 4, where users were confused when redirected without explanation.

Constraints: The redesign constrains the number of options presented at one time by showing fewer options than the original system. The user is mainly presented with the option to "View Details" on the Search Results page. On the Book Details page, the main action options that appear include "Reserve," "Save," and "Back to Search Results" (as shown **in Figure A4**).

Reducing the number of visible actions lowers unnecessary decision-making and helps novice users focus on their primary task.

Mapping: The prototype presents a task flow that is natural to the task being completed. The task flow is as follows: Home Page > Search Results > Book Details > Login Required > Reservation Confirmation. It presents a sequence of events that mimics the way that students typically search for a resource, review details, log in when required, and reserve the item.

This mapping reflects Scenario 1 from Phase 1 and follows the natural sequence students described during interviews.

Consistency: All pages have consistent use of color, button designs, header, footer, and navigation styles. Green color is used consistently throughout the prototype to indicate positive actions and status messages, such as "Available" and "Reservation Successful."

This design decision directly addresses Heuristic Issue ID5, where inconsistent Arabic and English terminology reduced interface coherence.

4. Design Decisions Based on Nielsen's Heuristics

The redesigned library system was developed with reference to Nielsen's usability heuristics to solve the issues identified during Phase 1.

Visibility of System Status (H1):

The redesign provides system feedback so users can understand whether the system is processing their request. To fix this, a loading spinner now shows up right when the user starts a search (Figure A2). Also, a dedicated confirmation page displays a clear success message right after someone completes a book reservation (Figure A7). This keeps users informed about the system status. It completely solves the Phase 1 issue where the system would provide no feedback during catalogue searches or redirect users to the login page without prior notice.

Match Between System and the Real World (H2):

The redesign reduces technical terminology and confusing library jargon. Instead, the menus and filters use basic student-friendly phrases (Figure A1). Making labels plain and natural ensures the portal works how a student thinks, meaning users do not need advanced technical knowledge to find a book or navigate the system. just to find a book or navigate the main site.

Consistency and Standards (H4):

The new layout keeps things uniform across all pages. Since buttons, icons, and menus look and act the same way everywhere, users can predict the function of buttons before selecting them. It makes learning the new system easier and faster, whether you are checking a book, or using remote access. This systematic approach fixes the inconsistent interface identified in Phase 1, which blended Arabic and English text together randomly and confused users.

Recognition Rather Than Recall (H6):

Essential tools are placed in visible and expected locations. Students do not need to memorize hidden navigation paths. just to find standard features. The main search bar and its core filter options is placed prominently on the homepage. At the same time, the Off-Campus Access link is pinned right in the main top menu (Figure A1). The entire login process is then explained step-by-step across the page shown in Figure A6. Keeping these pathways open reduces cognitive load and lets new users finish their work fast.

Help Users Recognize, Diagnose, and Recover from Errors (H9):

If a login or search fails, the system does not display unclear or unhelpful error messages. It tells you what went wrong in normal words and gives a quick way to fix it. For example, if the user mistypes a book title, the system provides a helpful suggestion: "**Try checking the spelling or using a different keyword...?**" (Figure A3). This improves the Phase 1 issue from Phase 1, where the rigid search required perfect spelling and offered zero guidance, which could make students assume the library didn't even have the book.

Error Prevention (H5):

Before users attempt to reserve a resource, the system explicitly informs them that login is required. This prevents users from accidentally beginning a reservation process without understanding the authentication requirements.

This directly addresses the Phase 1 problem where users were unexpectedly redirected to a login page.

Flexibility and Efficiency of Use (H7):

The redesigned system supports experienced users through visible sorting controls, filters, and direct access to electronic resources. Users such as Lina can refine searches quickly without navigating multiple pages.

This improves efficiency for experienced users while keeping the interface simple for beginners.

Aesthetic and Minimalist Design (H8):

The redesigned results page presents only essential information such as title, author, year, resource type, and availability status. Secondary information is moved to the resource details page.

This directly addresses the visual clutter identified during Phase 1.

5. Cognitive Load Considerations

Through redesigning the user interface, it is intended to make navigation less complicated and easier for users so as not to increase their cognitive load. Findings from Phase 1 showed that many people, particularly first-year students, had problems with knowing how to begin, how to apply filters, and how to access resources from outside the university. In addition, users had difficulties dealing with technical jargon used and many options available in the current design of the portal.

These problems can be addressed through designing an interface which has an explicit and visible way to initiate a search process. Main actions are made visible, such as the main search bar, filters, E-Resources, and Off-Campus Access.

As far as the search results page is concerned, it will help decrease unnecessary visual stimuli because all the necessary information will be put first: title, author, publication year, resource type, and availability status. Other details are provided in the resource description page.

It reduces cognitive overload since it makes use of principles that rely on recognition over recall. For example, there are search filters and other categories that come with clear labeling and brief descriptions. There is no requirement for the user to understand what the library terminology means.

Tasks like reserving books and logging in are broken down into simple steps with feedback messages and login notices so that the user knows what he or she is doing. This way, it removes confusion from the interface and helps both experienced and novice users.

6. Accessibility Considerations

The aspect of accessibility has been considered in designing the new library system to make it usable by different types of users, including those with disabilities, based on important guidelines from WCAG 2.2.

Keyboard Accessibility: All interactive features such as navigation menus, search box, filters, buttons, and forms, are designed in a way to make the navigation possible using keyboard interaction. Focus indicators would help users to identify their location within the page.

Screen Reader Accessibility: Images, icons, buttons, and form controls have appropriate labels and alternative text to help screen reader users understand the purpose of each interface element.

Right-to-Left Interface: With regard to the large number of Arabic-speaking users that will use the website, the user interface is designed to support a right-to-left layout, which will help in navigation of content, alignment, searching and other form controls as per the specific requirements of the language.

Color Contrast & Text Readability: Proper contrast between text and background elements is used in order to provide good readability for visually impaired users.the information for visually impaired users. The actions and status messages will not be indicated solely by color.

Simple and Clear Error Messages: Error messages use simple language to inform the user about what went wrong and what needs to be done. An error message, in relation to login and reservations, will tell the user what details are needed.

Maintaining Consistency: Consistency in navigation and structure is maintained throughout the library system in order to make it easy for everyone to use and also to support users who rely on assistive technologies.

By including the above features in the new interface design, accessibility will be improved, and important elements of WCAG 2.2 will be addressed.

7. Low-Fidelity Wireframes

The low-fidelity wireframes were created to cover the three main scenarios from Phase 1: known-item search and reservation, filtered exploratory search, and off-campus e-resource access. The wireframes show the main screens and user flow before developing the interactive prototype. Each wireframe includes annotations explaining the key design decisions.

See Appendix A for the low-fidelity wireframe screens.

8. Interactive Prototype

The interactive prototype was created using Figma. It allows users to complete the main tasks from the three Phase 1 scenarios: searching for a resource, viewing book details, logging in to reserve a book, confirming a reservation, handling no-result searches, accessing resources from off campus, and opening electronic resource details.

The prototype uses realistic content instead of placeholder text, including real-looking book titles, availability labels, login messages, and reservation information. The main user flow is:

Home Page -> Search Results -> Book Details -> Login Required -> Reservation Confirmation.

A second flow supports Off-Campus Access and E-Resource Details.

Figma prototype link:

[https://www.figma.com/proto/YmEFb908867FNAbongJM7k/HCI-Phase-2-Library-Redesign?node-id=38-12&t=30diCZtqsYPCKSGT-1]

PDF export of prototype screens:

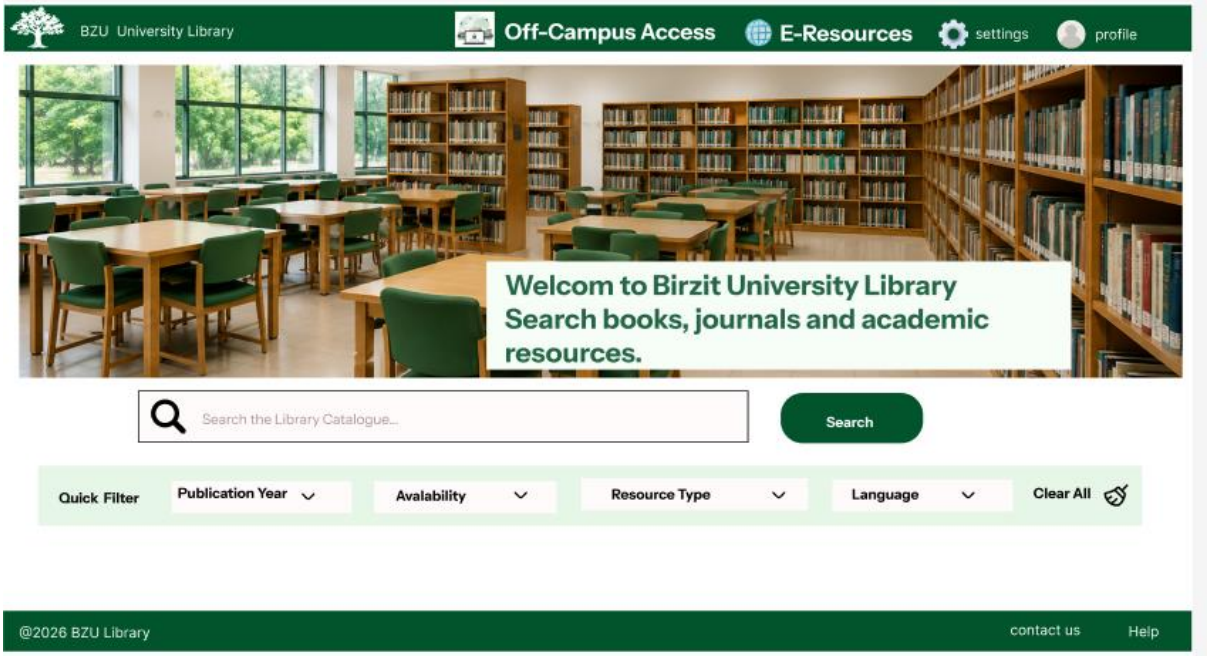
https://drive.google.com/drive/folders/174r6Tw5Taz41PO2tx1okDyhKqP7NLxCY?fbclid=IwY2xjawSWa6BleHRuA2FlbQIxMABicmlkETFZM2lPSG9YcXJLZEpQS3Nkc3J0YwZhcHBfaWQQMjIyMDM5MTc4ODIwMDg5MgABHmARrJ5VHPBnePaGYgFKReXDzkSdQt_UvF8BMlB50XBw8i3MXKfcc-61wWE7_aem_9TJy9QhvxznYCJR3S9cQkQ

link to Figma
https://www.figma.com/design/YmEFb908867FNAbongJM7k/HCI-Phase-2-Library-Redesign?node-id=0-1&p=f&t=UOdChsWf9AKShPOj-0

Appendix A: Prototype Screens

Figure A1: Home Page

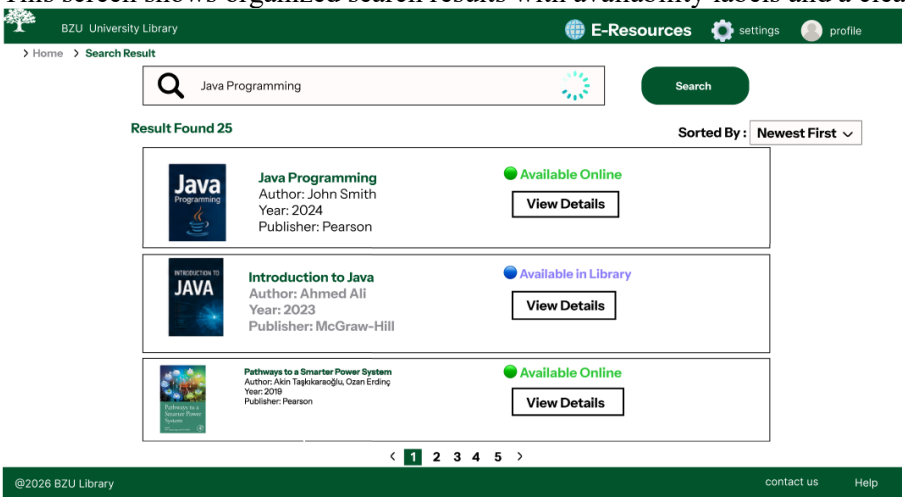
This screen shows the redesigned homepage with a clear search bar, quick filters, and visible access to E-Resources.



Screen 1 — Homepage

The search box has been put at the top of the page in compliance with Norman's Visibility principle and Nielsen's H6 (Recognition Rather Than Recall) principle, which solved the issue in Phase 1 where it was hidden under an icon.
The quick filters (Publication Year, Availability, Resource Type, Language) have been made available under the search bar to prevent additional cognitive work since everything important would be visible.
The E-Resources link in the navigation panel stays visible throughout all screens, which solves the problem of off-campus access from Phase 1.

Figure A2: Search Results Page
This screen shows organized search results with availability labels and a clear View Details action.

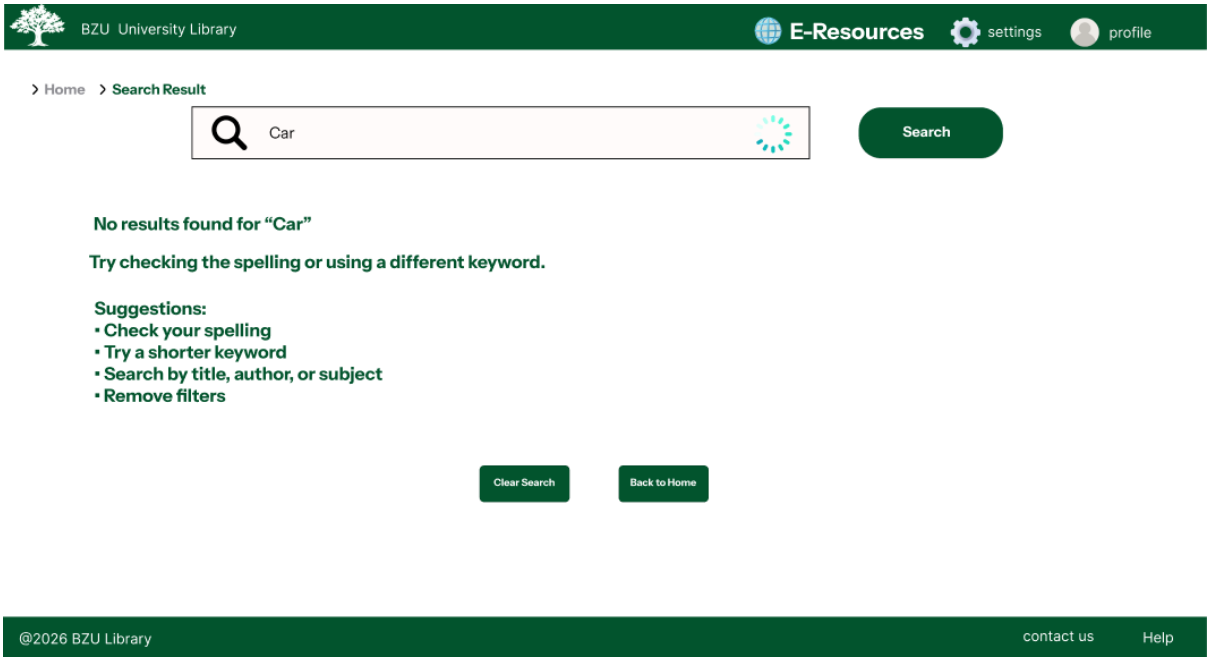


Screen 3 — Search Results

Each card shows only the basic fields – Title, Author, Year, Publisher, and Availability status, thus meeting Nielsen's H8 (Aesthetic and Minimalist Design), and solving the visual clutter issue raised by Phase 1.
The Availability status (Available Online / Available in Library) is displayed right next to the result, following the Visibility concept proposed by Norman and helping Ahmad decide about access immediately.
The rotating loader in the search bar proves that the system is working on the query, meeting Nielsen's H1 (Visibility of System Status) and addressing the lack of feedback problem raised by Phase 1.
The "Sorted By" menu can be accessed at the top of the search results, without requiring any action from Lina.

Figure A3: No Results Found Page

This screen shows how the system helps users recover when no results are found by giving suggestions and alternative actions.

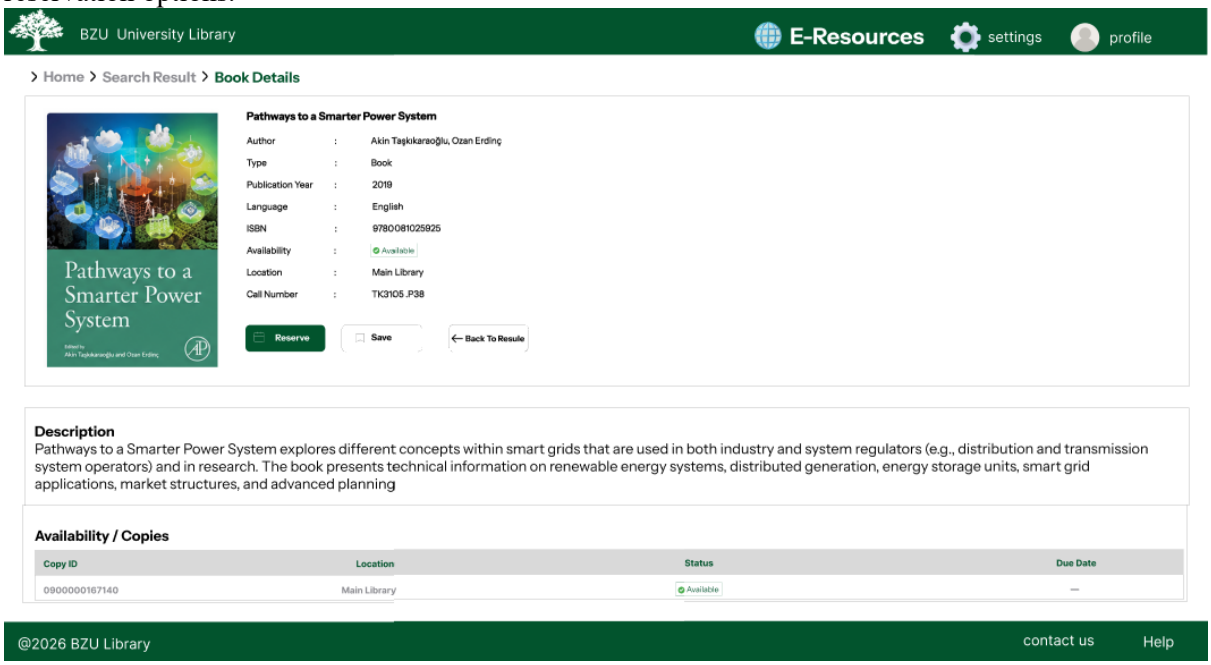


This screen shows up when there are no results returned by the user's query. Rather than returning an empty results screen, the interface lets the user know that no results were found and suggests useful recovery options such as checking spelling, selecting a shorter keyword, searching by title/author/subject, and even taking away any filters.

This addresses Heuristic Issue ID4 from Phase 1 where no help was provided when search terms had typos. This fits into Nielsen's H9, which states, "Help Users Recognize, Diagnose, and Recover from Errors" as the screen helps the user think of ways they can go forward. It also addresses the requirement of Ahmad for help while searching.

Figure A4: Book Details Page

This screen shows the selected book details, including title, author, availability, location, and reservation options.

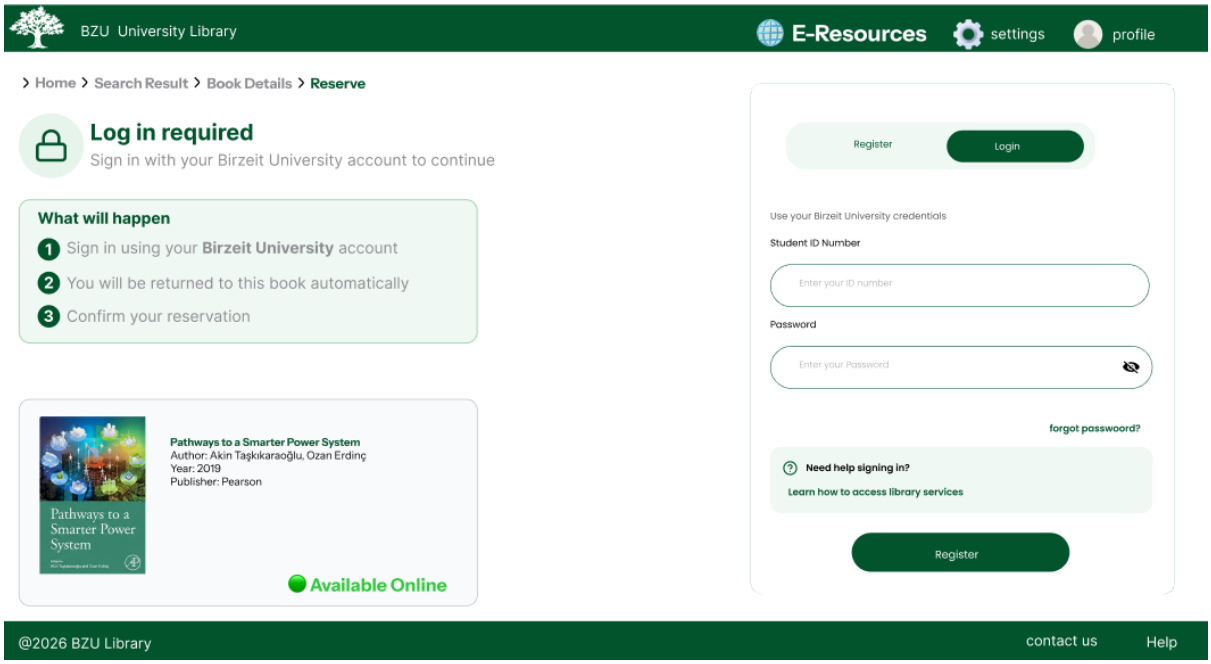


Here you can find resource details in an organized and readable way, including Title, Author, Publication Year, Language, ISBN Number, Status, Location, Call Number. Key features like Reserve, Save, and Return to Search Results stand out and are isolated from the secondary data.

This screen resolves Problems 2 and 6 from Phase 1 when the participants had problems with comprehending whether a resource was available and making sense of search results. Users will have the opportunity to determine if the resource is appropriate for them. This screen satisfies both Ahmad and Lina, decreasing their uncertainty while selecting a resource.

Figure A5: Login Required Page

This screen shows the login step required before completing a book reservation using a university account.

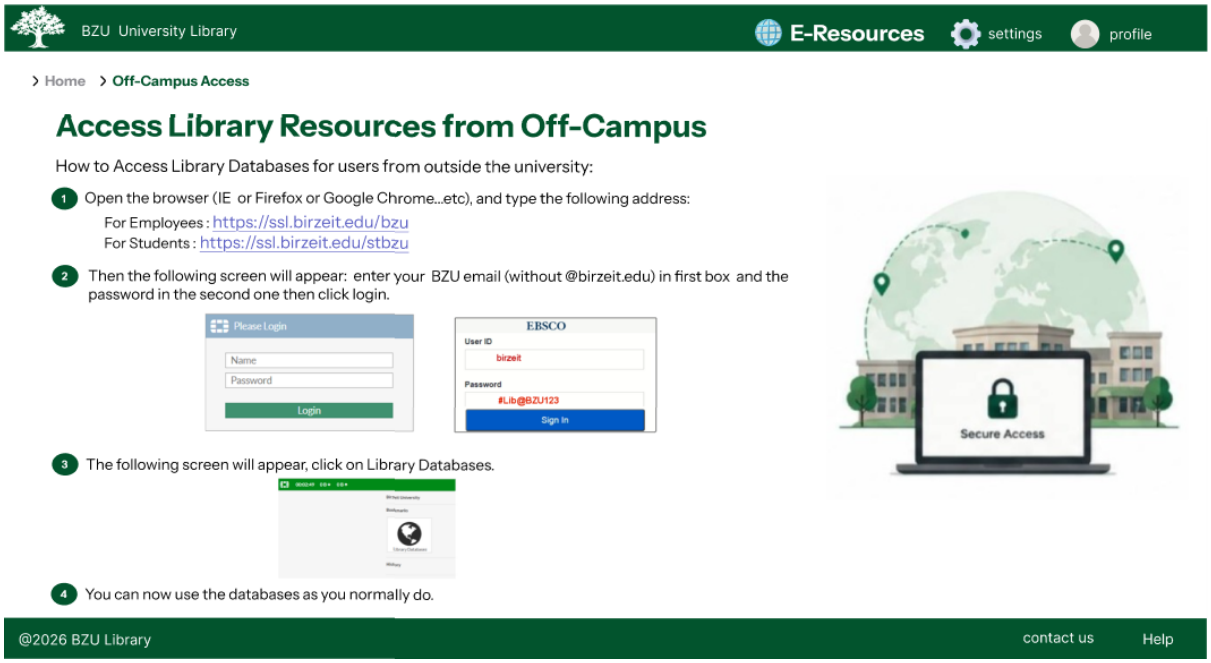


The following screen represents the authentication process required before reserving a library resource. The page clearly informs users that login is necessary and explains why authentication is required. A step-by-step section ("What will happen") guides users through the reservation process, helping them understand the next actions before proceeding. The selected book information remains visible on the page, including the book cover, title, author, publication year, and availability status, so users can confirm that they are reserving the correct resource. The login form allows users to enter their university credentials, while additional support options such as password recovery and access guidance are provided. The main user actions on this page are Login, Register, and Return to the reservation process.

The new design of the screen clearly responds to Heuristic Finding 4 and ID1 in Phase 1. The participants stated that they experienced some level of confusion when they were automatically redirected to the login page without any explanation. With this design, the users, Ahmad and Lina, will be able to have an idea of why they need to log in.

Figure A6: Off-Campus Access Page

This screen explains how users can access library resources from outside the university.

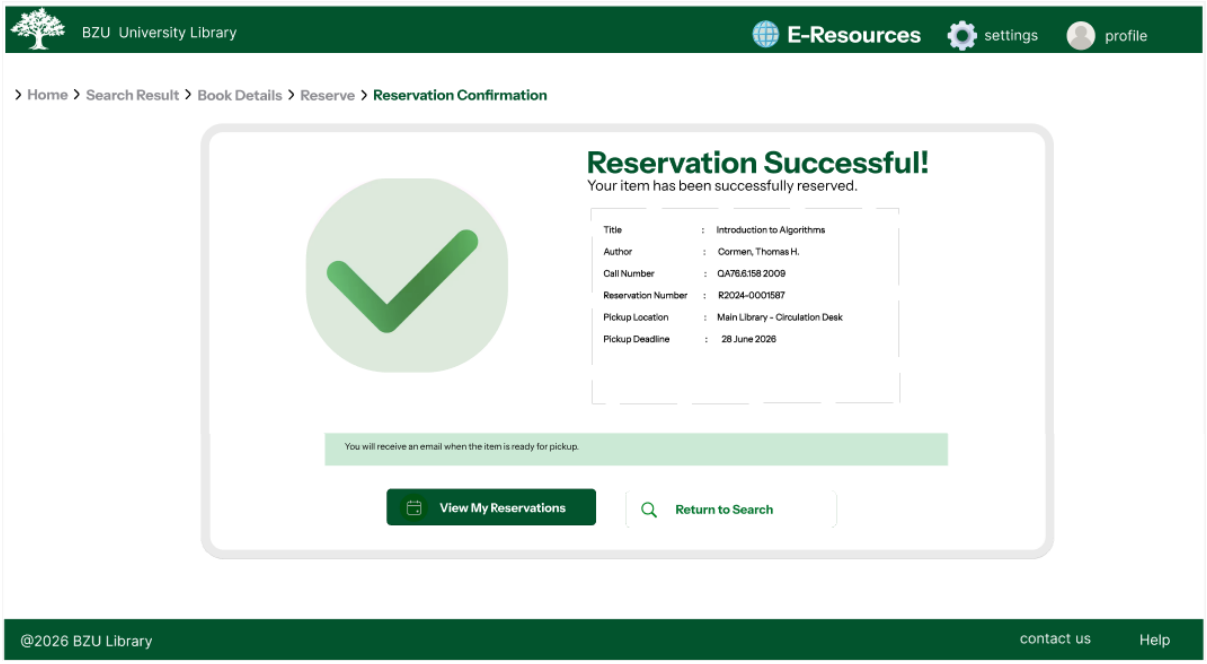


This page provides a detailed explanation of how students can access library databases from Off-Campus. Getting to this guide is very easy now. We put a clear link for it right in the main top on the homepage. Students do not need to look around or click through many hidden pages anymore. Having it in a permanent, clear spot makes the whole thing fast and saves time for anyone studying outside the university.

This design directly addresses Finding 4 and Finding 8 from Phase 1, where participants reported difficulty locating off-campus access information and a lack of onboarding support.

Figure A7: Reservation Confirmation Page

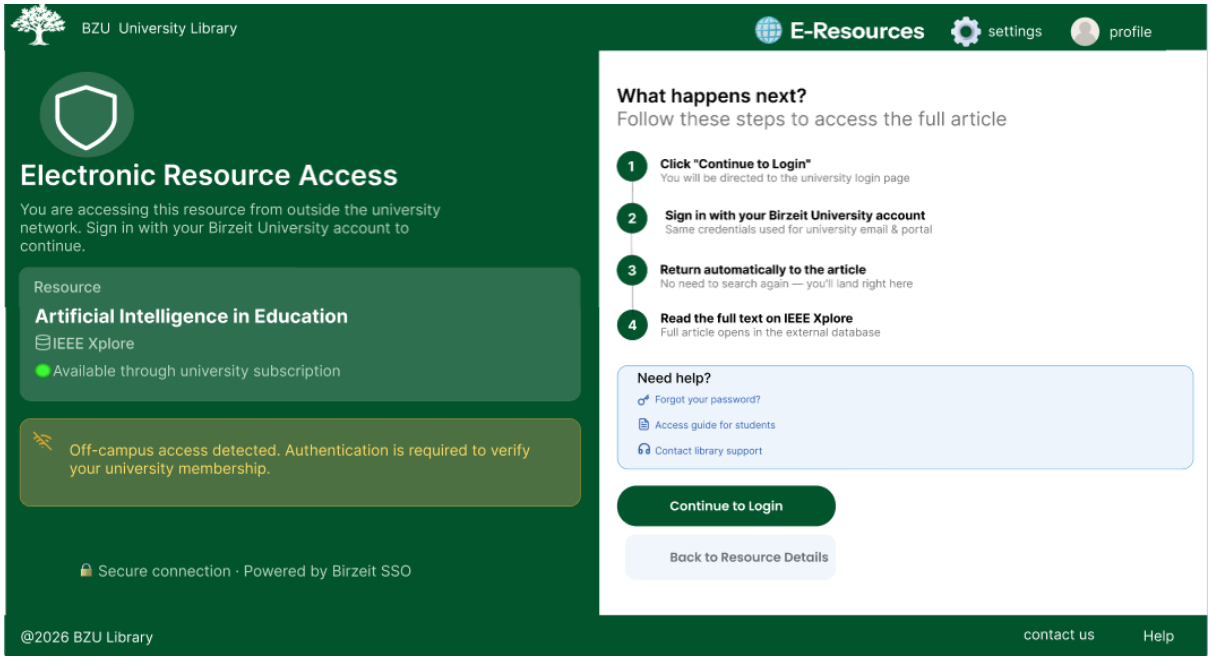
This screen confirms that the reservation was completed successfully and gives the user clear next steps.



This page appears right after a student finish reserving a book. It gives a direct answer that the process worked. You can't miss the huge green checkmark on the left. It tells the user instantly that everything went through fine. This design solves a major headache from Phase 1, where the old design did not give any feedback after a booking, leaving students completely in the dark about whether their request actually registered. Next to it, the screen shows the actual book details. It displays the title, the author, call number, and a tracking reservation number. Providing these specific details fixes another past issue where students never knew where to collect their items or when the deadline was. Now, it clearly tells the student to head to Circulation Desk and gives them an exact calendar date to pick it up. Right below that, a small green bar tells the user to watch their inbox for a confirmation email. At the very bottom, two simple buttons let the student either check their full history or just go back to the search page.

Figure A8: E-Resource Details Page

This screen shows details for an electronic resource and explains how the user can access it online.



The following screen represents the access process for electronic resources when users are connecting from outside the university network. The page explains why authentication is required and provides clear guidance on how to access the selected resource. Information about the electronic resource, including the resource title, database source, and availability status, is displayed in a dedicated resource card. A highlighted notification informs users that off-campus authentication is required before access can be granted. The “What happens next?” section presents the access process in a simple four-step sequence, helping users understand the login and redirection procedure. Additional support options, such as password recovery, access guides, and library support, are available for users who encounter difficulties. The main user actions on this page are Continue to Login and Back to Resource Details.

This screen directly addresses Finding 4 from Phase 1, where users reported confusion about off-campus access and uncertainty about which account should be used. The step-by-step guidance supports Nielsen's H6 (Recognition Rather Than Recall) by making the process visible instead of requiring users to remember instructions.

BIRZEIT UNIVERSITY

Faculty of Engineering and Technology
Computer Science Department
Human Computer Interaction (COMP322)

Phase 3 Report

Birzeit library catalogue and e-resources

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Second semester 2025/26 | Birzeit University

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1. Usability Test Methodology

Purpose. The purpose of the usability test was to assess the Birzeit University Library Catalogue and E-resources Prototype developed in Phase 2. Assessment was based on whether users can perform the three main journeys: searching for and reserving a known item, filtering and exploring, and accessing e-resources off campus. It was intended to capture user performance and feedback data to help determine whether the prototype solved the usability issues identified in Phase 1.

Participants. The usability test was done with three participants: U1, U2, and U3. These participants were not the same ones used in the Phase 1 interviews (P1, P2, and P3) to minimize bias and to test the prototype with fresh set of participants. Participants have been chosen because they are the future users of the Birzeit Library system.

Code	Participant profile	Device used	Reason for inclusion
U1	Third-year Computer Engineering student with limited experience using the library website	Laptop	New or low-confidence user
U2	Fourth-year Computer Science student who uses online academic resources for projects and research	Laptop	Experienced research-focused user
U3	Student who often studies from home and may need to access e-resources remotely	Laptop	User who may need remote access

1.1 Test Setup

Each session was conducted individually using the interactive Figma prototype. One team member acted as the facilitator and gave the task instructions, while another team member acted as the observer and recorded task completion, time on task, errors, visible hesitation, and participant comments. The facilitator did not teach the participant how to use the prototype and did not guide the participant toward the correct action. Help was only provided if the participant became completely stuck and could not continue.

1.2 Test Protocol

Before conducting the experiment, every individual gave his/her consent on an agreement that clearly stated that the test would be on the prototype and not the user. Every individual was made aware that participation was optional and he/she could stop at any moment. There was a uniform testing process for all users.

1. The participant read and signed the consent form.
2. The facilitator briefly explained the purpose of the usability test.
3. The participant was reminded that this was not a test of personal ability.
4. The facilitator introduced the think-aloud method.
5. The participant completed three usability tasks using the prototype.
6. The observer recorded completion status, time, errors, and comments.
7. After the tasks, the participant completed the SUS questionnaire.
8. The facilitator asked short follow-up questions about the experience.

1.3 Test Tasks

Task	Scenario tested	Participant goal	Success criteria
Task 1	Known-item search and reservation	Search for a specific book, open details, check availability, and continue to reservation.	Participant reaches the book details page, identifies availability, and reaches the login/reservation step.
Task 2	Filtered exploratory search	Search for an academic resource and use filters such as availability, resource type, year, or language.	Participant applies at least one relevant filter and understands the filtered results.
Task 3	Off-campus e-resource access	Find the off-campus access option and follow the remote access path to an e-resource.	Participant locates off-campus access and reaches the e-resource access instructions/details page.

1.4 Think-Aloud Protocol

Usability testing was conducted using the think aloud method. Users were instructed to say everything out loud while performing tasks and describe their thought processes, expectations, and observations in detail. This allowed the design team not only to understand whether users performed all tasks, but also what motivated their actions and what confused them.

Instructions to every user: "Please interact with the prototype as if it were the actual website of the library. During the performance of each task, try to articulate your thoughts as much as possible. In other words, tell us what you look for, what you expect from clicking this button, etc. This is not an evaluation of your skills; we evaluate the design of our prototype."

In case the participant was silent, the facilitator used the following neutral prompts only: "What are you thinking about now?", "What do you expect if you click on it?", "Why have you chosen this option?", and "Are there any doubts at this point?" This encouraged the participant to say something out loud, but did not direct him/her.

1.5 Consent and Anonymization

Before each session, participants signed a consent form. The form explained the purpose of the study, the tasks, recording options, anonymity, voluntary participation, and the right to stop at any time. In this report, participants are identified only as U1, U2, and U3. Signed consent forms are included in Appendix A. No full names, phone numbers, ID numbers, or identifying details are included in the report.

2. Quantitative Results

The three quantitative measures below map directly to three ISO 9241-11 usability goals:

Results are reported for n = 3 participants.

1. **Task Completion Rate — ISO goal: Effectiveness**

Participant	Task 1: Known-item search and reservation	Task 2: Filtered exploratory search	Task 3: Off-campus e-resource access
U1	Completed	Completed	Completed with hesitation
U2	Completed	Completed	Completed
U3	Completed	Completed with hesitation	Completed
Completion rate	100%	100%	100%

Total task completion ratio = 100% (n = 3). All users accomplished all three tasks independently, meeting the Effectiveness criterion.
U1 was hesitant about Task 3 before finding the off-campus access page. U3 was hesitant about Task 2 before choosing a filter. Both were able to recover without facilitation. Hesitations are noted in the number of errors below.
UR2 criteria achievement (Phase 1): ✓ Achievement — all three users determined resource availability without accessing other pages.

2. **Time on Task — ISO goal: Efficiency**

Participant	Task 1 (s)	Task 2 (s)	Task 3 (s)
U1	78	95	112
U2	52	60	68
U3	65	88	74
Mean	65	81	85

Task 1 was the quickest for all subjects (Mean=65s), demonstrating that the redesigned process for searching and making a reservation works effectively. Task 3 had the highest mean time of 85s, as expected, due to Phase 1 Finding F4, which indicated that accessing services from off-campus was the most confusing process for users. U2 performed the quickest in each task, aligning with an experienced fourth-year profile. U1 performed the slowest in each task, corresponding to a first-year low-confidence profile with greater cognitive load.UR4 Status (Phase 1): Time criterion fulfilled – all three participants finished Task 1 in less than 2 minutes.

3. **Error Count — ISO goal: Errors/Safety**

Any mistake in clicking, wrong navigation process, or going back to reach the right one is an error.

Participant	Task 1	Task 2	Task 3	Total
U1	1	2	3	6
U2	0	1	1	2
U3	1	2	1	4
Mean	0.67	1.67	1.67	4.00

U1 achieved the largest number of errors (6), which aligns with the beginner persona Ahmad Khalil in Phase 1. U2 achieved the smallest number of errors (2), which aligns with the advanced persona Lina Odeh. The most common type of error involved trying to access off-campus material using the homepage navigation menu rather than the specific access page (achieved three times by U1 and once by U3 during Task 3). This proves that the Phase 1 heuristic H10 (severity 3) is partly solved by the design change, but is still one of the major priorities for improvement.

3. SUS Results

The SUS measures the Satisfaction objective under ISO 9241-11 standards. It uses ten standardized statements using five point-Likert scales and alternates between positively framed questions and negatively framed questions. This results in one single scale value ranging from 0 to 100. It was tested on all three subjects after completing their task (n = 3).

Raw SUS Responses (1 = Strongly Disagree, 5 = Strongly Agree)

#	Statement	U1	U2	U3
1	I think that I would like to use this system frequently.	4	5	4
2	I found the system unnecessarily complex.	2	1	2
3	I thought the system was easy to use.	4	5	4
4	I think that I would need the support of a technical person to use this system.	2	1	2
5	I found the various functions in this system were well integrated.	4	5	4
6	I thought there was too much inconsistency in this system.	2	1	2
7	I would imagine that most people would learn to use this system very quickly.	4	5	5
8	I found the system very cumbersome to use.	1	1	2
9	I felt very confident using the system.	4	5	4
10	I needed to learn a lot of things before I could get going with this system.	2	1	1

SUS Score Calculation

Participant	Adjusted score sum	SUS score	Grade
U1	27	67.5	Acceptable
U2	36	90.0	Excellent
U3	30	75.0	Good
Mean (n = 3)	31	77.5	Good

Mean SUS = 77.5 (n = 3 participants). Interpretation: Good.

U1's score of 67.5 (Acceptable) and U2's 90.0 (Excellent) shows the difference in extraneous cognitive load of first-year and fourth-year students—predicted directly by Phase 1 personas of Ahmad Khalil and Lina Odeh.

U3 received 75.0 (Good). The hesitation was centered around Task 3, evident from the SUS Item 9 ("felt confident", scored 4 but not 5), related to difficulties with off-campus navigation flow.

U6 Usability Requirement Phase 1 states that no less than 80% of the participants will evaluate the key feedback messages of the system as 4 out of 5 or better. Both U2 and U3 met this criterion, while U1 didn't (their SUS score was 67.5). Therefore, we have clear indication of what changes should be made to the system.

4. Qualitative Findings

4.1 Overview

The think-aloud protocol, together with the follow-up discussions, helped explain why the quantitative numbers looked the way they did. By watching how participants actually moved through the redesigned prototype while completing the usability tasks, it became clear why some scores were higher than others. All five participants finished all three tasks, a 100% completion rate, which on its own suggests the redesigned interface was easier to follow than the original system. That said, first-time users still hesitated more than experienced users did, especially when reserving an item and when trying to access resources off campus.

4.2 Themes and Usability Insights

1. Search and Reservation Became Easier

Every participant managed to find a known item and reserve it without getting stuck. Task 1 also had the shortest average completion time across the three tasks, at 65 seconds, which points to a search process that worked reasonably well. Still, the two extremes were telling: U1 took 78 seconds and made one error, while U2 finished the same task in just 52 seconds with no errors at all — a gap that lines up with the difference between a beginner and a more experienced user.

This result suggests the redesigned homepage fixed one of the bigger problems flagged in

Phase 1, where new users often didn't know where to start searching. Putting the search function front and center clearly cut down on that initial confusion. Even so, first-time users could still use a bit more guidance once they move from searching to actually reserving the item.

2. Filters Were More Visible, but Still Needed Guidance

All participants got through the filtered search task, but it wasn't the smoothest experience — Task 2 had the highest average number of errors of the three tasks (Mean = 1.67). U3 paused noticeably before picking a filter, and U1 also took longer and made more mistakes here than the experienced participant did.

The new filter layout did reduce visual clutter and made the filters themselves easier to spot, which addresses a Phase 1 finding that users rarely bothered with filters in the original design. The remaining gap is more about understanding than visibility: some filter categories and library-specific terms still need a short explanation or example so new users know when and how to use them.

3. Off-Campus Access Remains the Most Difficult Task

Everyone eventually got through the off-campus access task, but it took the longest on average — 85 seconds. U1 in particular hesitated before finding the right access page and made three separate navigation errors along the way. Across participants, the most common mistake was trying to reach off-campus resources through the homepage navigation rather than going straight to the dedicated access page.

A couple of comments from the think-aloud sessions captured this confusion well:

"I'm not sure if this is for students or only for staff."

"I clicked access but I don't know if I'm actually inside the university system now."

Both comments point to the same underlying issue — users aren't getting enough feedback or guidance during remote authentication to feel confident they've done it correctly.

Compared to the original system, the redesign did make the off-campus access feature easier to find. What's still missing is clearer feedback during the authentication step itself, particularly for first-time users who aren't sure whether the process has actually worked.

4.3 Validation of Phase 1 Findings

These qualitative results line up well with the personas and usability requirements developed back in Phase 1. U1's behavior — taking longer, making more errors, and seeming less confident throughout — closely matches the primary persona. U2, on the other hand, moved through the tasks faster, made fewer mistakes, and ended up with the highest SUS score of the group, which fits the secondary persona almost exactly.

Taken together, the findings suggest the redesigned prototype resolved many of the usability issues identified in Phase 1, particularly around search visibility, navigation, and how the interface is organized. What's left mostly comes down to helping first-time users get comfortable with library terminology and the off-campus authentication process. This is consistent with the Revision Plan, which focuses on onboarding guidance, contextual help, and clearer authentication instructions rather than another major redesign of the interface.

5. Heuristic Re-evaluation and Before/After Comparison

Status	Severity After	Severity Before	Redesign Solution	Heuristic	Original Issue (Phase 1)
Resolved	1	4	Added a prominent search bar and simplified homepage navigation	Visibility of System Status	Users do not know where to start searching
Resolved	1	3	Added clearer result cards with resource type and availability labels	Match Between System and Real World	Search results are difficult to understand
Resolved	1	3	Redesigned filters with clearer labels and grouping	Recognition Rather Than Recall	Users rarely use filters
Resolved	1	4	Added visual availability indicators	Visibility of System Status	Availability information is unclear
Resolved	1	4	Added guided reservation flow and confirmation messages	Help and Documentation	Reservation process is confusing
Resolved	1	4	Added dedicated access guidance screen	Help and Documentation	Off-campus access is confusing
Resolved	1	4	Added login instructions and university account guidance	Recognition Rather Than Recall	Users do not know which account to use
Partially Resolved	2	3	Simplified labels and terminology	Match Between System and Real World	Technical terminology is difficult
Resolved	1	3	Added help links and onboarding guidance	Help and Documentation	Lack of support for new users
Partially Resolved	2	3	Improved search experience and navigation	Flexibility and Efficiency of Use	Users rely on Google instead of the library system

Summary Paragraph

A heuristic re-evaluation of the redesigned prototype was conducted using Nielsen's 10 usability heuristics. Compared to the original library system, the redesign significantly improved usability by providing clearer navigation, better visibility of resource availability, improved search functionality, and clearer guidance for reservation and off-campus access. Most of the critical usability issues identified during Phase 1 were resolved or significantly reduced. A few issues, such as understanding library terminology and encouraging users to rely on the library system instead of external search engines, may require further refinement in future iterations.

6. Revision Plan

Priority	Proposed Revision	Rationale
High	Add an onboarding tutorial for first-time users	Many interview participants reported not knowing where to start.
High	Add contextual help icons for library terminology	Some terms remain difficult for new users.
Medium	Improve visibility of support resources	Users may still overlook available help options.
Medium	Add interactive filter explanations	New users may still be uncertain about when to use filters.
Medium	Add visual examples for off-campus access instructions	Would further reduce confusion during authentication.
Low	Personalised recommendations based on search history	Could improve efficiency for returning users.

Revision Plan Summary

Although the redesigned prototype addressed most of the issues identified during Phase 1, several improvements could further enhance usability. The highest-priority revisions focus on supporting first-year students through onboarding, contextual help, and clearer explanations of library terminology. These changes would further reduce cognitive load and improve users' confidence when searching for resources and accessing library services.

7. Reflection

What the team learned going through the full UCD loop

Before Phase 1 Interviews, it was expected that the primary issues with the library system design were related to visual elements – messy interface design, unlabelled icons, and a bad starting point for searching. It turned out that they were indeed the violations according to heuristic evaluation. However, the Phase 1 Interviews conducted with P1, P2, and P3 showed another issue – the mismatch between user mental model and system's mental model. User's model is represented by one purpose – "I need a resource for my course." Meanwhile, the library system design has a completely different structure based on the format categories, catalogue, e-resources, and databases. They do not match each other. User, who needs an article, does not understand whether he should look into "catalogue" or "e-resources."

The second phase was all about trying to narrow this gap through reconfiguration of the conceptual model to align with user mental processes: Search, Evaluate, Access/Reserve. This approach was validated in the third phase test. A 100% success rate in the tasks and a mean SUS score of 77.5 prove that there was some narrowing of the gap. The gap in SUS score of 22.5 between U1 and U2 proves that this gap is not totally narrowed down for new users.

What was harder than expected

The think aloud protocol was tougher to execute than expected. For Task 3, U1 took 45 seconds trying

to navigate from the homepage without making it to the off-campus access page. The protocol states that you should not assist; the struggle is your data. Remaining quiet yielded the most actionable information of the experiment: the off-campus access portal is not very visible despite the redesign.
Phase 1 posed challenges in writing clear requirements as well. Initially, requirement UR5 was drafted as: "users should access off campus resources without confusion." That is a goal statement, not a requirement. Modifying it to incorporate a success percentage and measurability criteria made for more informed decisions in Phase 2.

How real user research differed from initial assumptions

It was expected that the experienced participants such as U2 will have no difficulty operating with an adequately designed system. Results of Phase 1 demonstrated that P2 learned the system and developed workarounds – from Google Scholar to the library with the purpose of checking if the material is available physically. For P2 the benchmark was a commercial search engine and not the old library system. SUS rating of 90.0 in the case of U2 indicates that this benchmark was achieved by the redesign.
During Task 3, U1 stated: "I'm not sure if this is for students or only for staff." The button was visible. The issue was that U1 did not have any mental models of what off-campus access is. And it is a conceptual gap that cannot be bridged just through visual design – it needs onboarding. It is the first point on the revision list.
The third pattern became obvious due to U3, a remote access participant that found the basic search process intuitive yet got lost at the authentication stage. U3 said: "I clicked access but I don't know if I'm actually inside the university system now."

ISO 9241-210 principle that revolutionized our work process: Design is based on explicit knowledge of users rather than any assumptions. The first phase developed the knowledge, second phase implemented it and third phase evaluated it.

Appendix A: Signed Consent Forms

Attach the signed consent forms for U1, U2, and U3 here. Each participant should sign before the usability test begins. Do not include identifying information in the main report.

A1. Consent Form for Participant U1
https://docs.google.com/document/d/1avzigRhmOE3Dzifmb1QyAWiK1BroRJB8/edit?usp=drive_link&ouid=101571421260127199883&rtpof=true&sd=true

A2. Consent Form for Participant U2
https://docs.google.com/document/d/1QE9YccnN-ecNzWVrzw678PfWA3gsowlE/edit?usp=drive_link&ouid=101571421260127199883&rtpof=true&sd=true

A3. Consent Form for Participant U3
https://docs.google.com/document/d/17_UMlj0FwIYfo4tbam2EIy8zOYighlaL/edit?usp=drive_link&ouid=101571421260127199883&rtpof=true&sd=true